

Systematic Options Strategies on the S&P 500: Buy-Write, Protective Put, Collar, and Long Straddle

Jean Philippe N'Dri

Quantitative Researcher | 2026

I. DATA

A. Dataset

We use a daily end-of-day SPY options dataset covering January 2020 to December 2022 (782 trading days). The data source is Kaggle. Each row of the dataset represents the information associated with one strike price and a given expiration date on a given quote date. Specifically, each row contains the bid, ask, implied volatility, and delta for both the call *and* the put at that strike and expiry.

B. Key Columns

Table 1. Key columns in the options dataset.

Column	Description
QUOTE_DATE	Quotation date
UNDERLYING_LAST	SPY closing price
EXPIRE_DATE	Option expiration date
DTE	Days to expiration
STRIKE	Strike price
C_DELTA / P_DELTA	Call / put delta
C_BID, C_ASK	Call bid and ask
P_BID, P_ASK	Put bid and ask
C_IV / P_IV	Call / put implied volatility

II. OPTIONS STRATEGIES: CONSTRUCTION

A. Roll Engine

At each roll date, options are selected by a two-step algorithm: **(1)** find the expiry whose DTE is closest to the target DTE; **(2)** within that expiry, select the strike whose delta is closest to the target delta. All trades execute at mid-price. Roll parameters for all four strategies are summarised in Table 2.

Table 2. Roll parameters for each strategy.

Strategy	Legs	DTE	Δ	Freq.
Buy-Write	$-C$	7	+0.50	Weekly
Prot. Put	$+P$	30	-0.30	Monthly
Collar	$+P, -C$	30	± 0.25	Monthly
Long Straddle	$+C, +P$	30	± 0.50	Monthly

B. Buy-Write (Covered Call)

The buy-write holds one unit of SPY and sells one ATM call each week. The P&L at expiration is the sum of three components: the change in the stock position, the premium received, and the settlement of the short call.

Full payoff at expiration (premium received c , strike K_c , entry price S_0):

$$\Pi_{BW}(S_T) = \underbrace{(S_T - S_0)}_{\text{stock P\&L}} + \underbrace{c}_{\text{premium}} - \underbrace{\max(S_T - K_c, 0)}_{\text{short call settlement}} \quad (1)$$

Evaluating the two cases explicitly:

$$\Pi_{BW}(S_T) = \begin{cases} S_T - S_0 + c & S_T \leq K_c, \\ K_c - S_0 + c & S_T > K_c. \end{cases} \quad (2)$$

Maximum gain is capped at $K_c - S_0 + c$; the break-even is $S_0 - c$.

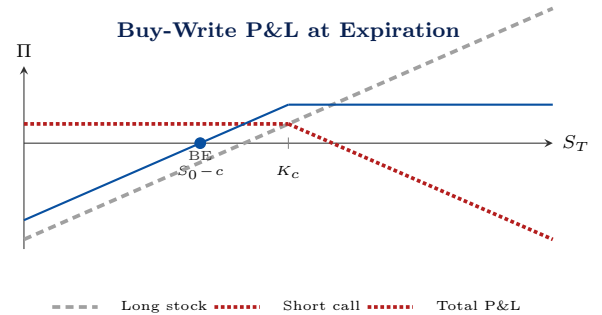


Figure 1. Buy-write: long stock + short call. Cap at $K_c - S_0 + c$, break-even at $S_0 - c$.

C. Protective Put

The protective put holds one unit of SPY and buys one OTM put each month, establishing a floor on portfolio value. The P&L at expiration combines the stock move, the premium paid, and the settlement of the long put.

Full payoff at expiration:

$$\Pi_{PP}(S_T) = \underbrace{(S_T - S_0)}_{\text{stock P\&L}} - \underbrace{p}_{\text{premium}} + \underbrace{\max(K_p - S_T, 0)}_{\text{long put settlement}} \quad (3)$$

Collecting terms:

$$\Pi_{PP}(S_T) = \begin{cases} K_p - S_0 - p & S_T < K_p, \\ S_T - S_0 - p & S_T \geq K_p. \end{cases} \quad (4)$$

Maximum loss is bounded at $K_p - S_0 - p$; upside is unlimited net of the premium drag.

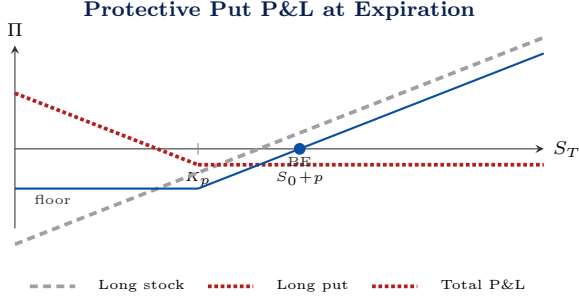


Figure 2. Protective put: long stock + long put. Floor at $K_p - S_0 - p$, break-even at $S_0 + p$.

D. Collar

The collar augments the protective put with a short OTM call, using the call premium to offset the cost of the put.

Full payoff at expiration (put premium p , call premium c , net cost $\delta = p - c$):

$$\begin{aligned} \Pi_{COL}(S_T) = & \underbrace{(S_T - S_0)}_{\text{stock P\&L}} - \underbrace{\delta}_{\text{net premium}} \\ & + \underbrace{\max(K_p - S_T, 0)}_{\text{long put}} - \underbrace{\max(S_T - K_c, 0)}_{\text{short call}} \end{aligned} \quad (5)$$

Evaluating the three cases explicitly:

$$\Pi_{COL}(S_T) = \begin{cases} K_p - S_0 - \delta & S_T < K_p, \\ S_T - S_0 - \delta & K_p \leq S_T \leq K_c, \\ K_c - S_0 - \delta & S_T > K_c. \end{cases} \quad (6)$$

Both profit and loss are bounded.

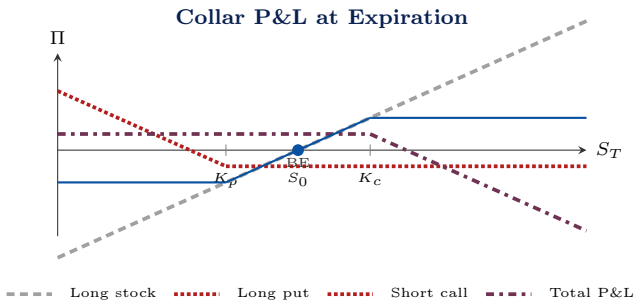


Figure 3. Collar: long stock + long put + short call. Floor and cap both bounded.

E. Long Straddle

The long straddle buys both the ATM call and the ATM put with no underlying position. It is a direction-neutral bet that the realized move will exceed the implied move.

Full payoff at expiration:

$$\begin{aligned} \Pi_{LS}(S_T) = & \underbrace{\max(S_T - K, 0)}_{\text{long call}} + \underbrace{\max(K - S_T, 0)}_{\text{long put}} \\ & - \underbrace{(c + p)}_{\text{total premium}} \end{aligned} \quad (7)$$

Since exactly one of the two max terms is positive (or both are zero when $S_T = K$), this simplifies to:

$$\Pi_{LS}(S_T) = |S_T - K| - (c + p) \quad (8)$$

Break-even points are $K \pm (c + p)$.

The strategy is profitable if and only if $\sigma^{RV} > \sigma^{IV}$.

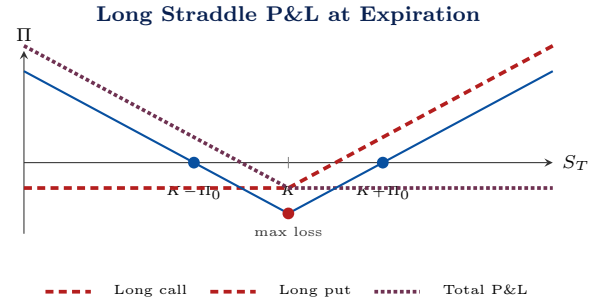


Figure 4. Long straddle: long call + long put. Max loss $-(c + p)$ at K , break-evens at $K \pm (c + p)$.

F. Daily Portfolio Valuation

Each strategy is represented as a daily index rebased to 100 at inception. The general form of the daily portfolio value V_t differs by strategy:

$$V_t^{BW} = S_t + \text{Cash}_t - C_t^{\text{MTM}} \quad (9)$$

$$V_t^{\text{PP}} = S_t + \text{Cash}_t + P_t^{\text{MTM}} \quad (10)$$

$$V_t^{\text{COL}} = S_t + \text{Cash}_t + P_t^{\text{MTM}} - C_t^{\text{MTM}} \quad (11)$$

$$V_t^{\text{LS}} = \text{Cash}_t + C_t^{\text{MTM}} + P_t^{\text{MTM}} \quad (12)$$

where C_t^{MTM} and P_t^{MTM} are the daily mid-price values of the open call and put positions respectively.

G. Cash Account

The cash account Cash_t accumulates all premium flows:

- **At roll:** cash receives (for sold legs) or pays (for bought legs) the mid-price premium.
- **At expiration:** cash is debited by the intrinsic value of any short leg that expires in-the-money, or credited by the intrinsic of any long leg.

Formally, for a short call at strike K_c :

$$\text{Cash}_{\text{expiry}} -= \max(S_T - K_c, 0) \quad (13)$$

and for a long put at strike K_p :

$$\text{Cash}_{\text{expiry}} += \max(K_p - S_T, 0) \quad (14)$$

H. Index Level

The strategy index is defined as:

$$I_t = 100 \times \frac{V_t}{V_0} \quad (15)$$

where V_0 is the portfolio value on the first trading day. For comparison, the SPY buy-and-hold index is:

$$I_t^{\text{SPY}} = 100 \times \frac{S_t}{S_0} \quad (16)$$

Rebasing both series to 100 at $t = 0$ allows direct visual comparison of cumulative returns regardless of the absolute dollar value of the initial portfolio.

I. Settlement and Roll Timing

Expiration and roll occur on the *same* trading day:

1. Settle the expiring position using the end-of-day SPY price.
2. Immediately sell (or buy) the new option using end-of-day mid-prices on the same date.

This same-day roll is operationally realistic: SPY weekly options expire at the open on Friday, and a new position can be opened before the 16:00 close. The settlement condition uses strict equality:

$$\text{current_expiry} == \text{date} \quad (17)$$

to avoid double-settlement on days when the position is already closed.

J. Look-Ahead Bias

The backtest is free of look-ahead bias. All prices used on date t are sourced exclusively from the end-of-day snapshot of date t : option selection, execution, and mark-to-market all use the same 16:00 cross-section. The only modest optimism is mid-price execution, which assumes fills exactly at the midpoint of the bid-ask spread.

III. PERFORMANCE METRICS

Let $\{r_t\}_{t=1}^n$ denote the daily returns of strategy index I_t , with $r_t = I_t/I_{t-1} - 1$.

A. Return and Volatility

$$\mu_{\text{ann}} = \left(\prod_{t=1}^n (1 + r_t) \right)^{252/n} - 1 \quad (18)$$

$$\sigma_{\text{ann}} = \hat{\sigma}(r) \cdot \sqrt{252} \quad (19)$$

B. Risk-Adjusted Ratios

The Sharpe ratio measures excess return per unit of total volatility:

$$\text{Sharpe} = \frac{\mu_{\text{ann}}}{\sigma_{\text{ann}}} \quad (20)$$

The Sortino ratio penalises only downside deviations, making it more appropriate for strategies with asymmetric return distributions:

$$\text{Sortino} = \frac{\mu_{\text{ann}}}{\sigma_{\text{down}} \cdot \sqrt{252}} \quad (21)$$

where $\sigma_{\text{down}} = \text{std}(r_t \mid r_t < 0)$.

The Calmar ratio relates annualised return to the maximum loss ever sustained:

$$\text{Calmar} = \frac{\mu_{\text{ann}}}{|\text{MaxDD}|} \quad (22)$$

The Omega ratio measures the ratio of gains to losses relative to a threshold (here zero), capturing the full shape of the distribution:

$$\text{Omega} = \frac{\sum_{r_t > 0} r_t}{\left| \sum_{r_t < 0} r_t \right|} \quad (23)$$

C. Drawdown

The drawdown at date t measures the percentage decline from the running peak:

$$\text{DD}_t = \frac{I_t - \max_{s \leq t} I_s}{\max_{s \leq t} I_s} \quad (24)$$

The maximum drawdown is $\text{MaxDD} = \min_t \text{DD}_t$.

D. Tail Risk

Value-at-Risk at the 95% confidence level gives the loss threshold exceeded on only 5% of days:

$$\text{VaR}_{95\%} = -\text{percentile}(r, 5\%) \quad (25)$$

Conditional Value-at-Risk (CVaR), also known as Expected Shortfall, is the average loss on the worst 5% of days:

$$\text{CVaR}_{95\%} = -\mathbb{E}[r \mid r \leq -\text{VaR}_{95\%}] \quad (26)$$

CVaR is a coherent risk measure (?) and is particularly informative for option strategies whose return distributions exhibit significant skewness and excess kurtosis.